

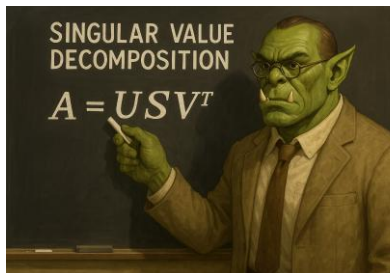
New Models, Services, and Frameworks

Native image generation with ChatGPT

We had only a couple of weeks to appreciate Google's take on it, and now OpenAI strikes us with their own effort :)

It's quite likely that OpenAI had this model for quite some time already (I've heard about it in mid-24 or even earlier), but they were reluctant to show it. But now, it's available when you chat with GPT-4o.

Generally, better than Gemini, and it has some boons, even though its images are more somber and less imaginative than of DALLÉ-3. First of all, it generates text on images surprisingly well. Its editing capability is also surprisingly great.



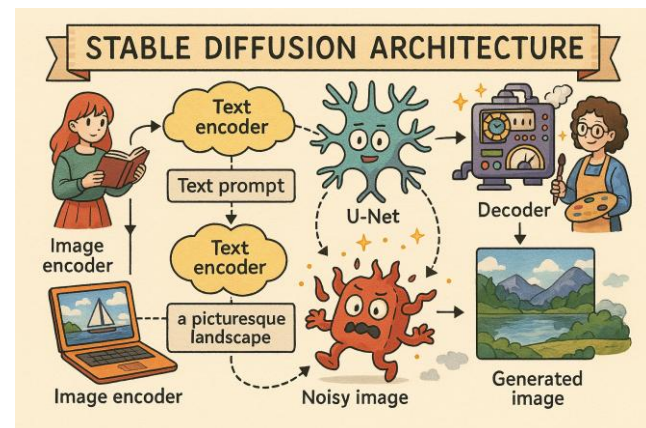
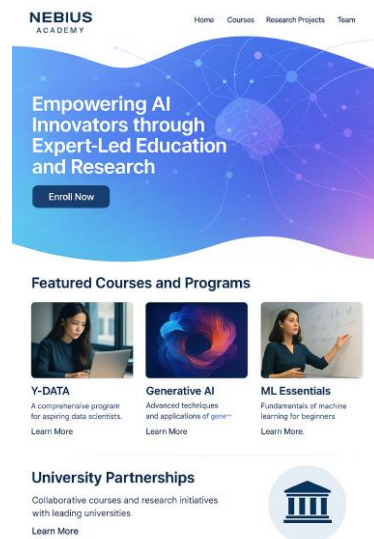
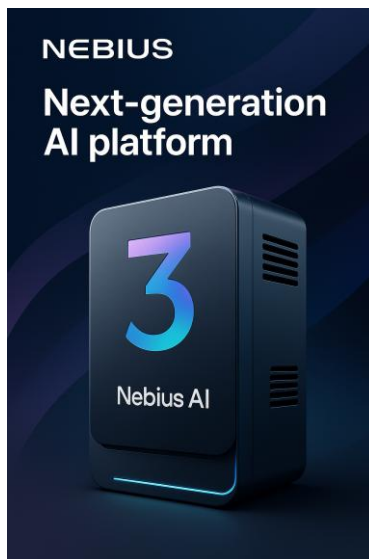
“Make them wear brimmed hats with feathers”



And what positively excited me is that you can create arts of three characters and then ask to make a team of them – and it really works, even though some additional characters may appear as hallucinations:



Some more cool things :) The Stable Diffusion architecture isn't probably entirely wrong... The image is fun anyway



And, of course, I couldn't resist trying to turn a photo from our London party into anime. (Sorry, I know you're probably already tired of Ghibli-style generations)



Gemini 2.5

<https://blog.google/technology/google-deepmind/gemini-model-thinking-updates-march-2025/#enhanced-reasoning>

A new reasoning model from Google, which is, if we trust in benchmarks, very cool and becomes one of the best reasoning models. I've copy-pasted a table with numbers; it's on the next page. I myself didn't have much time to test it yet, but at least it solved a math problem for me with which GPT-4.5 failed.

As other Gemini models, this one also boasts million-scale context size – which, as we know, doesn’t actually mean that it’s able to “understand” such long contexts. However, it gets 90% score (the highest of them all) on the [fiction.live](#) benchmark, which is an assortment of long stories with questions such as

"He was a terrific assassin. He eliminated many Imperial soldiers, their names were...

Question: Finish the sentence, what names would Jerome list? Give me a list of names only.

(Maybe still sounds a little like search, but need to check more questions.)

Benchmark		Gemini 2.5 Pro Experimental (03-25)	OpenAI o3-mini High	OpenAI GPT-4.5	Claude 3.7 Sonnet 64k Extended Thinking	Grok 3 Beta Extended Thinking	DeepSeek R1
Reasoning & knowledge Humanity’s Last Exam (no tools)		18.8%	14.0%*	6.4%	8.9%	—	8.6%*
Science GPQA diamond	single attempt (pass@1)	84.0%	79.7%	71.4%	78.2%	80.2%	71.5%
	multiple attempts	—	—	—	84.8%	84.6%	—
Mathematics AIME 2025	single attempt (pass@1)	86.7%	86.5%	—	49.5%	77.3%	70.0%
	multiple attempts	—	—	—	—	93.3%	—
Mathematics AIME 2024	single attempt (pass@1)	92.0%	87.3%	36.7%	61.3%	83.9%	79.8%
	multiple attempts	—	—	—	80.0%	93.3%	—
Code generation LiveCodeBench v5	single attempt (pass@1)	70.4%	74.1%	—	—	70.6%	64.3%
	multiple attempts	—	—	—	—	79.4%	—
Code editing Aider Polyglot		74.0% / 68.6% whole / diff	60.4% diff	44.9% diff	64.9% diff	—	56.9% diff
Agentic coding SWE-bench verified		63.8%	49.3%	38.0%	70.3%	—	49.2%
Factuality SimpleQA		52.9%	13.8%	62.5%	—	43.6%	30.1%
Visual reasoning MMMU	single attempt (pass@1)	81.7%	no MM support	74.4%	75.0%	76.0%	no MM support
	multiple attempts	—	no MM support	—	—	78.0%	no MM support
Image understanding Vibe-Eval (Reka)		69.4%	no MM support	—	—	—	no MM support
Long context MRCR	128k (average)	94.5%	61.4%	64.0%	—	—	—
	1M (pointwise)	83.1%	—	—	—	—	—
Multilingual performance Global MMLU (Lite)		89.8%	—	—	—	—	—

Compute Optimal Scaling of Skills: Knowledge vs Reasoning

This paper strives to refine the **scaling laws** determining the balance between model size and data size at training, given some total amount of FLOPs. The authors investigate, whether the optimal balance depends on the task.

This might be not the most natural question (LLMs are trained with many tasks in mind); anyway, let's understand the technicalities.

Scaling laws research often operate with **Isoflop curves**, that is curves showing how some target metric changes as we vary parameter count (or data size) with total FLOPs fixed. On the two pictures below (left and central) you can see such Isoflop curves for knowledge-related tasks and for coding.

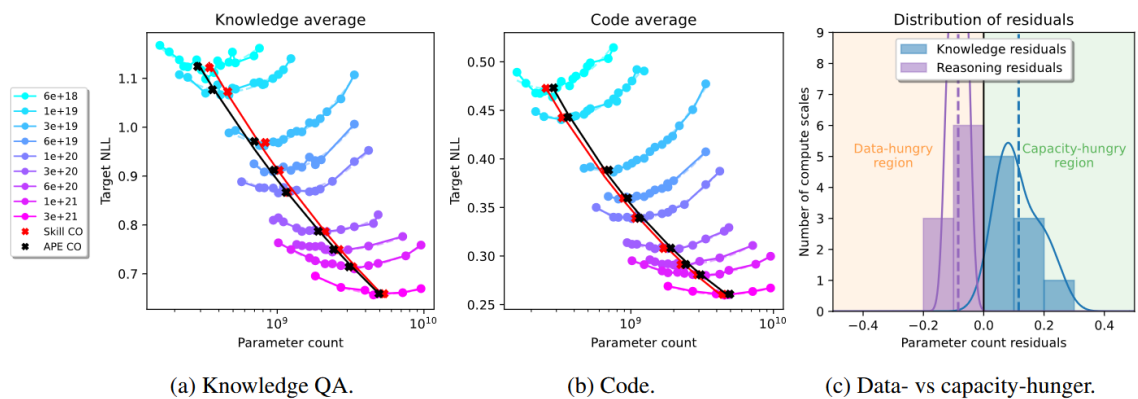


Figure 1: **Isoflop curves and COs for code and knowledge QA, along with the APE COs.** In (a), we see that on average across held-out datasets, knowledge QA tends to be capacity-hungry compared to the APE CO. On the other hand, in (b), we see that code tends to be data-hungry relative to the APE CO. We show the distribution of these relationships in (c), where we plot distributions of the log-scale differences in parameter count between skill-dependent COs and APE COs and find their means lie on opposite sides of the APE CO. The black curves in (a) and (b) represent the predicted APE COs from [Dubey et al. \(2024\)](#), mapped to their respective IsoFLOP groups.

The real question now is which metrics to choose. By default, we'd take just log likelihood, but now we need to relate the metrics to our task domains. A natural way of doing this is to choose some task-related benchmark and measure NLL (negative log likelihood) on it. That's which datasets they choose for each of the task domains:

Split	Knowledge skills					Code skills				
Hypothesis	Trivia QA (dev)	NQ (dev)	SQuAD (train)	MMLU (train)		RepoBench	BigCodeBench	CAT-LM (test gen.)	MultiPL-E (HumanEval)	MBXP
Held-out	Trivia QA (test)	Trivia QA (wiki test)	NQ (test)	SQuAD (dev)	MMLU (val)	SWE-bench (oracle)	BigCodeBench (test gen.)	CrossCodeEval	HumanEval	MBPP

Table 1: **Hypothesis and held-out splits for code and knowledge-based skills.** In all our experiments, we first develop hypothesis on a *hypothesis split* and then confirm them on a *held-out split* containing different datasets for the same skills. References to each of these datasets are provided in the text.

Now, let's return to the Isoflop curves above and understand the takeaways. A black diagonal line on each of the plots shows what the optimal balance would be without taking the tasks into account, while the red lines show task-specific optima. Now, this shows that:

- Optima for the knowledge are shifted towards larger parameter count,
- Optima for coding are shifted towards dataset size.

So, it looks like you need more parameters to make a model more knowledgeable, while it takes more data to make it more skillful.

Search-R1: Training LLMs to Reason and Leverage Search Engines with Reinforcement Learning

<https://arxiv.org/pdf/2503.09516>

This is quite a natural adaptation of the idea of budget forcing (which was initially used to stimulate long reasoning) to improve how the LLM works with search. In a sense, this results in a version of Deep Research.

The idea becomes quite clear from the prompt:

Answer the given question. You must conduct reasoning inside `<think>` and `</think>` first every time you get new information. After reasoning, if you find you lack some knowledge, you can call a search engine by `<search>` query `</search>`, and it will return the top searched results between `<information>` and `</information>`. You can search as many times as you want. If you find no further external knowledge needed, you can directly provide the answer inside `<answer>` and `</answer>` without detailed illustrations. For example, `<answer>` Beijing `</answer>`. Question: `question`.

Thus, during the generation process, several cycles of (initializing search – getting information – thinking again) pass until the model is able to give the final answer. On the next page there's a screenshot of an example of an output from the paper.

During training, the authors suggest masking retrieved information thus not taking them into account for the loss.

The model is trained with RL (not surprising). The authors only used the correct answer reward, not incorporating format rewards, because they turns out to be unnecessary: the model eventually copes with the format well just thanks to prompting.

The authors check this approach on several 3 to 7B LLMs and get quite good results.

I'm now looking forward to seeing some kind of scaling laws measuring the optimal between reasoning and retrieval. I'd wager that the result would be aligned with the takeaways of the previous paper: the more knowledge-related the task is, the more

important the search is, and in the case of purely-knowledge-related question, as we know, reasoning may even hurt.

Question: Curious is a women's fragrance by a singer born in what city and state?

SEARCH-R1:

<think> I need to find out which city and state a singer, by whom the women's fragrance "Curious" was created, was born in. </think>

<search> Curious fragrance information </search>

<information> Doc 1(Title: "Curious (fragrance)") Curious (fragrance) Curious is a women's fragrance by Britney Spears for Elizabeth Arden, and is the first perfume to be endorsed by Britney Spears, preceding ""Fantasy."" Spears reportedly earned \$52 million from the endorsement. ""Curious"" was released in September 2004, and was very successful internationally. It was the number one fragrance of 2004 in department stores, and in 2005, ""Curious"" was honored by the Fragrance Foundation as Best Women's Fragrance. The fragrance is a white floral scent. Its notes are Louisiana magnolia, golden Anjou pear, lotus flower, tuberose, star jasmine, pink cyclamen, vanilla-infused musk, sandalwood and blonde woods. Elizabeth Arden ... </information>

<think> I need to find out which city and state Britney Spears, the singer behind "Curious" fragrance, was born in. </think>

<search> Britney Spears birthplace </search>

<information> Doc 1(Title: "Britney Spears") the best-selling female albums artist of the 2000s. In 2004, she launched a perfume brand with Elizabeth Arden, Inc., from which sales exceeded US\$1.5 billion. . Spears was born in McComb, Mississippi, the second child of Lynne Irene Bridges and James Parnell Spears. Her maternal grandmother, Lillian Portell, was English (born in London), and one of Spears's maternal great-great-grandfathers was Maltese. Her siblings are Bryan James and Jamie Lynn. Britney was born in the Bible Belt, where socially conservative evangelical Protestantism is a particularly strong religious influence. Spears was baptized into the Southern Baptist Convention, but later in life studied </information>

<think> Since Britney Spears was born in McComb, Mississippi, I need to find out which city and state that is. </think>

<search> McComb, Mississippi location </search>

<information> Doc 1(Title: "McComb, Mississippi") McComb, Mississippi McComb is a city in Pike County, Mississippi, United States, approximately south of Jackson. As of the 2010 census, the city had a total population of 12,790. It is the principal city of the McComb, Mississippi Micropolitan Statistical Area. McComb was founded in 1872 after Henry Simpson McComb of the New Orleans, Jackson and Great Northern Railroad, a predecessor of the Illinois Central Railroad (now part of the Canadian National Railway), decided to move the railroad's maintenance shops away from New Orleans, Louisiana, to avoid the attractions of that city's saloons. The railroad purchased land in Pike County, </information>

<think> I have found that Britney Spears was born in McComb, Mississippi. Since the question asks for the city and state, I just need to confirm the state, which I already know from the previous search. </think>

<answer> McComb, Mississippi </answer>

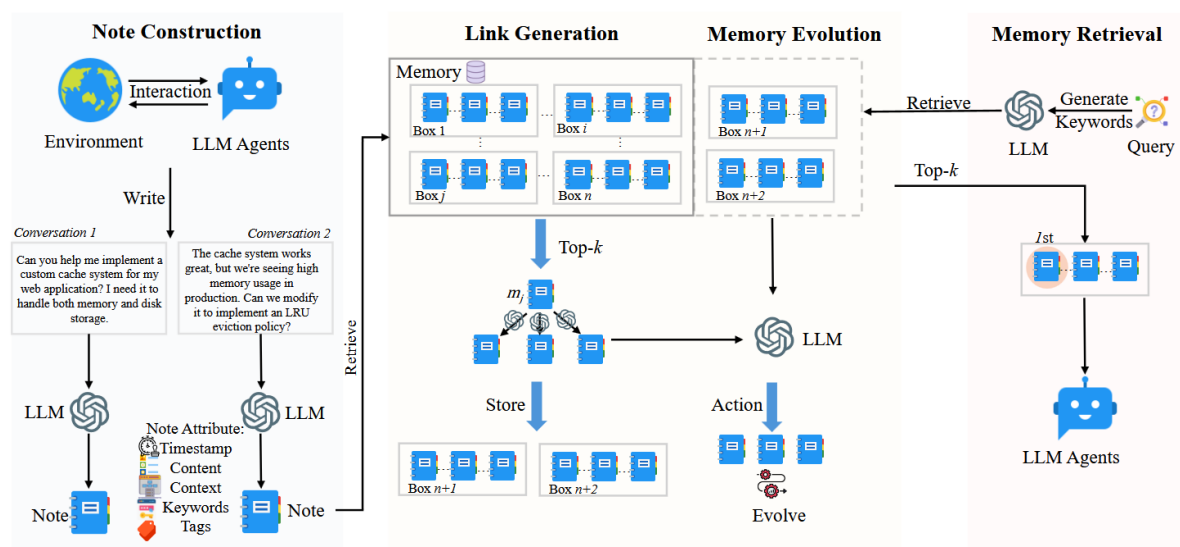
A-MEM: Agentic Memory for LLM Agents

<https://arxiv.org/pdf/2502.12110>

We hear much about RAG with vector stores, but to tell the truth just a vector store is a too simplistic model of memory, which may fall short in many cases. For example, I'm myself thinking much about RAG for math research, and I totally understand that only geometric proximity (and even only semantic proximity) isn't enough to find relevant things: you need to take into account connections between notions, implications between theorems and so much more! So, I was quite excited to check another paper about memory structuring.

In this paper, alas, the authors store not math theorems, but the history of human-agent interaction – which is very useful, of course; it's pretty annoying when an LLM gets confused about the details of your previous conversation.

The authors suggest not just a data storage, but a whole agentic system to manage memory. Let's understand how it works.



A single memory entry is a structured “**Note**”, which consists of

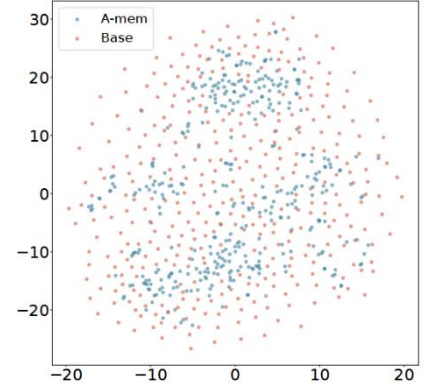
- the original interaction content,
- the timestamp of the interaction,
- LLM-generated keywords that capture key concepts,
- LLM-generated tags for categorization,
- LLM-generated contextual description that provides rich semantic understanding,
- the set of linked memories that share semantic relationships.

Now, the last one is interesting. The **links** are created in a two-step way, which is quite conventional these days:

1. First, the node is vectorized using some embedding model, and a number of geometrically nearest neighbours is fetched (cosine distance is used).
2. Then, an LLM is used to rerank them. Some top-k are chosen to receive connections with this note.

But that's not all! Now, each of these closest neighbours is **evolved** based on the new memory and the other closest neighbours – using an LLM, of course.

The **retrieval** in this system is solely distance-based, and that somewhat concerns me: I'm sure there should be a more interesting way of finding relevant things in such a structure. Still, the authors are able to show some metric improvements on the [LoCoMo dataset](#) containing quite long dialogs. Also, the geometry of the embedding space becomes more structured (see the picture on the right).



(b) Dialogue 2

DAPO: An Open-Source LLM Reinforcement Learning System at Scale

<https://arxiv.org/pdf/2503.14476>

The excitement about RL was recently renewed by the success of DeepSeek-R1. DeepSeek leveraged the novel GRPO algorithm instead of a more traditional PPO which was traditionally used for RLHF; however, later research showed that PPO may also work better (or may work worse, depending on the luck and savvy of those using it). Anyway, it's clear that RL is capricious and the AI community would only benefit from further investigation of RL algorithms.

This paper suggests an improvement of the GRPO algorithm (for the learning long reasoning task, of course). The usual GRPO looks like this:

Compared to PPO, GRPO eliminates the value function and estimates the advantage in a group-relative manner. For a specific question-answer pair (q, a) , the behavior policy $\pi_{\theta_{\text{old}}}$ samples a group of G individual responses $\{o_i\}_{i=1}^G$. Then, the advantage of the i -th response is calculated by normalizing the group-level rewards $\{R_i\}_{i=1}^G$:

$$\hat{A}_{i,t} = \frac{r_i - \text{mean}(\{R_i\}_{i=1}^G)}{\text{std}(\{R_i\}_{i=1}^G)}. \quad (4)$$

Similar to PPO, GRPO adopts a clipped objective, together with a directly imposed KL penalty term:

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}_{(q,a) \sim \mathcal{D}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \left(\min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip} \left(r_{i,t}(\theta), 1 - \varepsilon, 1 + \varepsilon \right) \hat{A}_{i,t} \right) - \beta D_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}) \right) \right], \quad (5)$$

where

$$r_{i,t}(\theta) = \frac{\pi_{\theta}(o_{i,t} \mid q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t} \mid q, o_{i,<t})}. \quad (6)$$

Let me refresh this for you a bit:

- The **relative advantage** $\hat{A}_{i,t}$ measures the deviation of the reward r_i of an i -th object in its batch (hence the “GR” for Group Relative in **GRPO**). Comparing to using pure reward r_i , it makes the reward signal less noisy by normalizing it. (In PPO, a critic model would be used with similar intentions.)
- The $r_{i,t}$ ratio (not to be confused with the reward r_i !) resembles the “internal reward model” introduced in DPO.
- The $\text{clip}(r_{i,t}(\theta); 1 + \varepsilon; 1 - \varepsilon)$ part (ε is typically small, around 0.1-0.2) is a kind of a regularization trying to prevent the policy from changing too rapidly, which could happen in the case of a large $r_{i,t}(\theta)$.
- The D_{KL} summand (Kullback-Leibler divergence) is also the regularization term, explicitly discouraging π_θ from getting far from π_{old} . The authors of this paper decided to get rid of this term, because while fine tuning an LLM for long reasoning, the distribution may indeed change significantly, so this regularization may be counterproductive.

The authors try to address several issues of GRPO, and for that they do quite nice ablation research (which actually motivated me to review this paper).

1. **Entropy collapse phenomenon:** the entropy of the policy decreases quickly as training progresses. One of the possible reasons is how clipping behaves differently with low and high $r_{i,t}$. If for a token o_i and a prompt q the value of $\pi_{old}(o_i|q)$ is small, say, 0.1, and ε is the typical 0.2, then $\text{clip}\left(\frac{\pi_\theta(o_i|q)}{\pi_{old}(o_i|q)}; 1 - \varepsilon; 1 + \varepsilon\right)$ behaves like on the plot on the right.

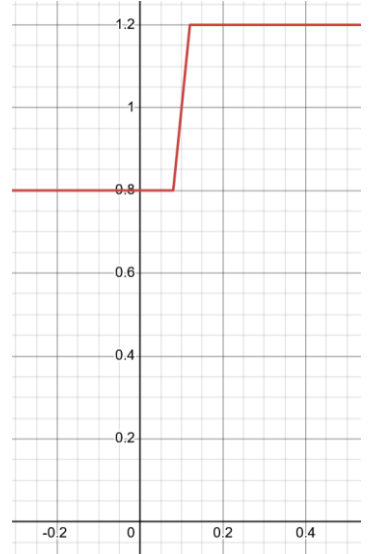
That means that the loss doesn’t care if $\pi_\theta(o_i|q)$ is $0.1 \cdot (1 + 0.2)$ or greater – and it wouldn’t push it to become greater. So, if a token was rare at the beginning, GRPO gives it “little motivation” to increase.

To account for this, the authors suggest using two different values ε :

$$\text{clip}\left(\frac{\pi_\theta(o_i|q)}{\pi_{old}(o_i|q)}; 1 - \varepsilon_{low}; 1 + \varepsilon_{high}\right)$$

Of course, with higher ε_{high} . This is called **Clip-Higher**.

2. **Optimization paralysis at rewards close to 1.** When the reward r_i (not $r_{i,t}$!), which is between 0 and 1, is getting so good that generally it’s close to 1, there is not much difference inside the batch, and thus $\hat{A}_{i,t}$ become close to 0.



The same holds true if the model is so bad that all the accuracies are around zero. So, the authors suggest **Dynamic Sampling** – filtering out “too successful tokens” from being counted in the loss during training.

3. **Low importance of individual tokens in long sequences for the training process.** Since in GRPO rewards are averaged over each sequence ($\frac{1}{|o_i|} \sum \text{reward for } o_i$), each individual token’s reward isn’t very important if the sequence is long enough. Because of that, the authors suggest global averaging over both tokens in a sequence and different sequences – they call it **Token-level loss**.
4. **Inadequate punishment of long outputs.** Though LLM reasoning may be very long, no training process can tolerate arbitrarily long sequences. So, after some max length they are cut short and (because the answer isn’t obtained), penalized. However, this may confuse the model, because it’s being scored for just length and not reasoning correctness. To account for that, the authors suggest to mask truncated sequences from loss computation and also to introduce **soft overlong punishment** that would discourage the model from generating longer sequences in a linear, more “understandable” way:

$$R_{\text{length}}(y) = \begin{cases} 0, & |y| \leq L_{\text{max}} - L_{\text{cache}} \\ \frac{(L_{\text{max}} - L_{\text{cache}}) - |y|}{L_{\text{cache}}}, & L_{\text{max}} - L_{\text{cache}} < |y| \leq L_{\text{max}} \\ -1, & L_{\text{max}} < |y| \end{cases}$$

The resulting objective is:

$$\begin{aligned} \mathcal{J}_{\text{DAPO}}(\theta) = & \mathbb{E}_{(q,a) \sim \mathcal{D}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \\ & \left[\frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip} \left(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}} \right) \hat{A}_{i,t} \right) \right] \\ \text{s.t. } & 0 < \left| \{o_i \mid \text{is_equivalent}(a, o_i)\} \right| < G, \end{aligned} \quad (8)$$

where

$$r_{i,t}(\theta) = \frac{\pi_{\theta}(o_{i,t} \mid q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t} \mid q, o_{i,<t})}, \quad \hat{A}_{i,t} = \frac{R_i - \text{mean}(\{R_i\}_{i=1}^G)}{\text{std}(\{R_i\}_{i=1}^G)}. \quad (9)$$

The main results, reflecting the contribution of each technique, is on the right. Feel free to check more plots in the paper!

I think, this analysis is quite insightful and helps to better understand what’s happening inside GRPO.

Table 1 Main results of progressive techniques applied to **DAPO**

Model	AIME24 _{avg@32}
DeepSeek-R1-Zero-Qwen-32B	47
Naive GRPO	30
+ Overlong Filtering	36
+ Clip-Higher	38
+ Soft Overlong Punishment	41
+ Token-level Loss	42
+ Dynamic Sampling (DAPO)	50

